



EAST WEST UNIVERSITY

LAB REPORT

Semester: Spring-2026

Course Name: Electronic Circuits

Course Code: CSE 251

Section No: 1

Group No: 10A

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Submission Date: 27/06/2026

Title:

To study the Performance of an Inverting and Non-Inverting Amplifier Using Op-Amp.

Objective:

1. To learn about the gain and performance of inverting and non-inverting amplifiers.
2. To learn how the operational amplifier works.
3. Gain hands-on experience with the basic configurations of inverting and non-inverting amplifiers using operational amplifiers (Op-Amps)

Theory:

An inverting amplifier is designed such that the input signal is applied to the inverting terminal of the op-amp through an input resistor, while the non-inverting terminal is grounded. A feedback resistor connects the output to the inverting terminal, creating a closed loop. The output signal of the inverting amplifier is the amplified version of the input signal but with an opposite phase, meaning it is inverted. The gain of the amplifier is determined by the ratio of the feedback resistor to the input resistor. This type of amplifier is commonly used in applications where signal inversion is required, such as summing amplifiers and integrators.

In contrast, a non-inverting amplifier applies the input signal to the non-inverting terminal of the op-amp, while the inverting terminal is connected to a voltage divider made of resistors. The output signal is an amplified version of the input signal and maintains the same phase as the input. The gain of the non-inverting amplifier is determined by the ratio of the resistors in the voltage divider and is always greater than or equal to one. Non-inverting amplifiers are often used when impedance matching is required or when a buffer is needed between stages of a circuit.

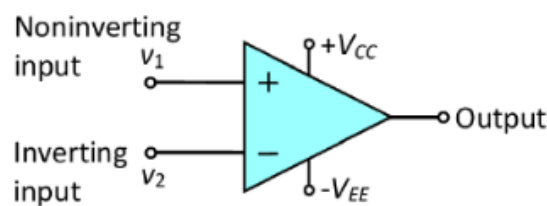


Figure 1: Electircal Symbol of Operation Amplifier

In the inverting amplifier, the input signal is applied to the inverting terminal through a resistor R_1 , while the non-inverting terminal is grounded. A feedback resistor R_f connects the output to the inverting terminal, ensuring negative feedback. The voltage gain of the amplifier is:

$$A_f = \frac{R_f}{R_1}$$

The negative sign shows that the output is 180° out of phase with the input.

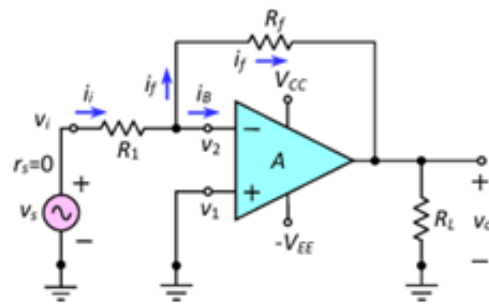


Figure 2: Inverting Amplifier

In the non-inverting amplifier, the input signal is applied to the non-inverting terminal, while the inverting terminal is connected through a voltage divider of R_1 and R_f . The gain is given by:

$$A_f = 1 + \frac{R_f}{R_1}$$

Here, the output is in phase with the input.

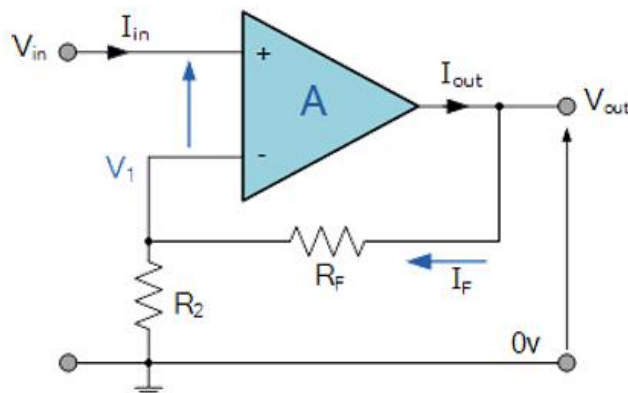


Figure 3: Non-Inverting Amplifier

Components and Apparatus:

1. Digital trainer board
2. Split DC supply
3. Signal generator
4. Oscilloscope
5. Digital multimeter
6. Op Amp: 741 (1 pc)
7. Resistor: 4.7K (2), 2.2K (1)
8. Connecting wire

Working Diagrams:

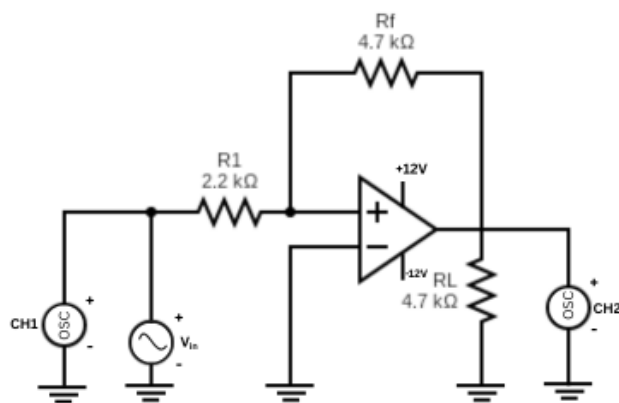


Figure 4: Inverting Amplifier

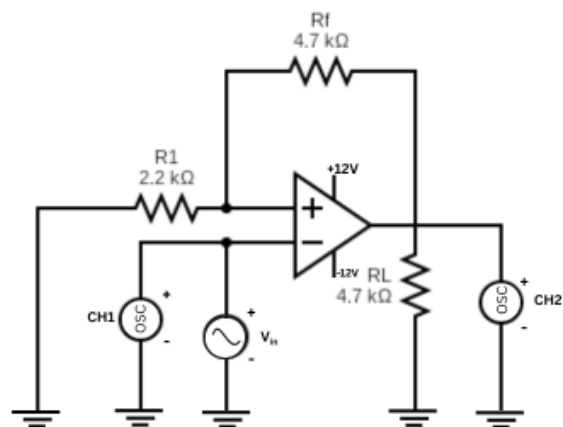


Figure 5: Non-Inverting Amplifier

Inputs and Outputs of Oscilloscope:

Inverting Amplifier:

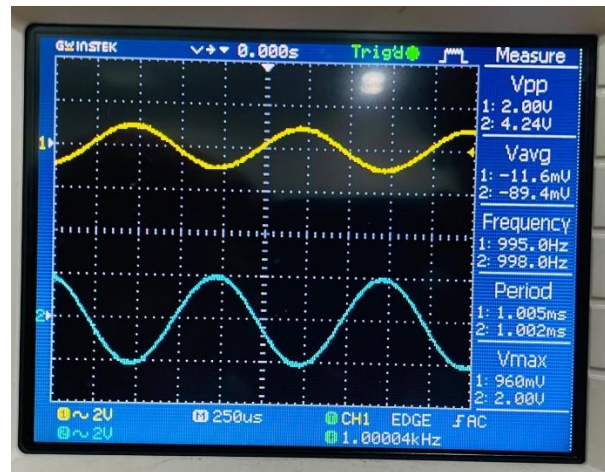


Figure 6: Oscilloscope Data for Inverting Amplifier

Input (CH1, yellow) and output (CH2, blue) waveforms measured using the oscilloscope. The output voltage (4.24 Vpp) is greater than the input voltage (2.00 Vpp), indicating signal amplification at approximately 1 kHz.

Non-Inverting Amplifier:

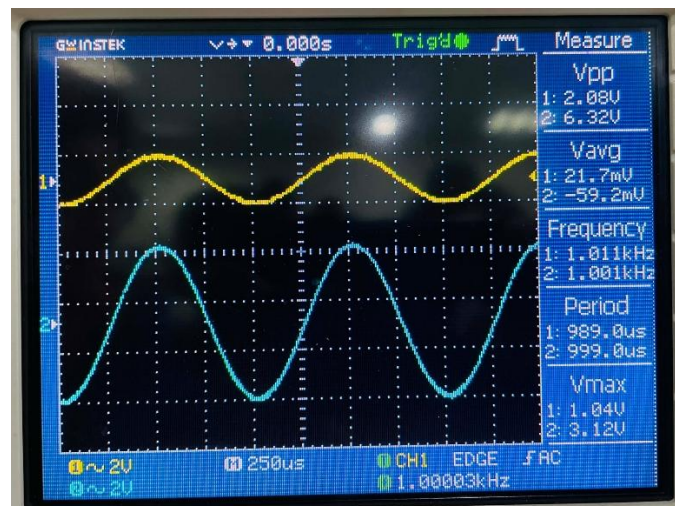


Figure 7: Oscilloscope Data for Non-Inverting Amplifier

Input (CH1, yellow) and output (CH2, blue) waveforms measured using the oscilloscope. The output voltage (6.32 Vpp) is approximately three times the input voltage (2.08 Vpp), confirming successful amplification while maintaining a frequency of approximately 1 kHz.

Calculation and Results

For Inverting Amplifier:

Theoretical gain:

$$\begin{aligned}A_f &= -\frac{R_f}{R_1} \\&= -\frac{4.7}{2.2} \\&= -2.13\end{aligned}$$

Practical gain:

$$V_{in} = 2V$$

$$V_{out} = 4.24V$$

$$\begin{aligned}\therefore A_f &= -\frac{V_{out}}{V_{in}} \\&= -\frac{4.24}{2} \\&= -2.105\end{aligned}$$

Error:

$$\begin{aligned}&\frac{\text{Theoretical gain} - \text{Practical gain}}{\text{Theoretical gain}} \times 100 \\&= \frac{2.13 - 2.105}{2.13} \times 100 \\&= 1.1\%\end{aligned}$$

For Non-Inverting Amplifier:

Theoretical gain:

$$\begin{aligned} A_f &= 1 + \frac{R_f}{R_1} \\ &= 1 + \frac{4.7}{2.2} = 3.13 \end{aligned}$$

Practical gain:

$$V_{in} = 2.08V$$

$$V_{out} = 6.40V$$

$$\begin{aligned} \therefore A_f &= \frac{V_{out}}{V_{in}} \\ &= \frac{6.40}{2.08} \\ &= 3.08 \end{aligned}$$

Error:

$$\begin{aligned} &\frac{\text{Theoretical gain} - \text{Practical gain}}{\text{Theoretical gain}} \times 100 \\ &= \frac{3.13 - 3.08}{3.13} \times 100 \\ &= 1.5\% \end{aligned}$$

Discussion:

In this experiment, the performance of both the inverting and non-inverting amplifiers was studied using a 741 operational amplifier. The circuits were built on a digital trainer board, and a 1 kHz sinusoidal input signal was applied using a signal generator. The input and output waveforms were observed with a digital oscilloscope, and the practical voltage gains were compared with the theoretical values calculated from the resistor ratios.

For the inverting amplifier, the theoretical voltage gain was -2.13 , while the practical gain obtained from the oscilloscope measurements was -2.105 . The output waveform was amplified and inverted with respect to the input signal, confirming the expected 180° phase shift of an inverting amplifier. The calculated percentage error was only 1.1%, indicating good agreement between theory and practical observation.

For the non-inverting amplifier, the theoretical gain was 3.13, whereas the practical gain was 3.08. The oscilloscope showed that the output signal was amplified while remaining in phase with the input signal. The percentage error was 1.5%, which also indicates that the circuit performed close to the expected theoretical value.

The small differences between the theoretical and practical results may be due to resistor tolerance, the non-ideal characteristics of the 741 operational amplifier, slight variations in the power supply, and minor measurement errors of the oscilloscope and signal generator. Despite these factors, the measured values were very close to the calculated values.

Overall, the experiment was successful and achieved its objectives. It provided a clear understanding of the working principles of inverting and non-inverting amplifiers and demonstrated how an operational amplifier can be used to amplify signals with different phase characteristics while maintaining the same input frequency.

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